

Differentiating the fluctuation function in its order:

$$\partial_\lambda S_\lambda(a) = \int_0^\infty (\log t) t^{\lambda-1} e^{-\beta t + \beta a \sqrt{t}} dt$$

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Abstract

Unit 11 of the grammar-paper §4 autoformalisation: the order-derivative identity `eq:log_lambda_derivative` at $k = 1$, $\partial_\lambda S_\lambda(a) = \int_0^\infty (\log t) t^{\lambda-1} e^{-\beta t + \beta a \sqrt{t}} dt$. Since $\partial_\lambda t^{\lambda-1} = (\log t) t^{\lambda-1}$, differentiating S_λ in its *order* inserts a $\log t$ under the integral; this is the analytic foundation of the paper’s log-insertion machinery (`lem:log_shift`, `prop:LogODE`, `lem:log_generating`), which encodes SLT pole multiplicities. Proved by differentiation under the integral sign, dominated by a $|\log t|$ -weighted fluctuation integrand whose integrability we establish. Zero `sorry/axiom`.

1 Setting

Units 1–10 differentiated S_λ in the *argument* a (the ladder). This unit differentiates in the *order* λ — a different analytic problem, because λ sits in the exponent of $t^{\lambda-1}$ and the derivative integrand acquires a $\log t$ factor that is singular at both ends of $(0, \infty)$. It is the last piece of §4 that is pure real analysis rather than operator algebra.

2 The things formalised

```
theorem abs_log_le_rpow_add (t : ℝ) (ht : 0 < t) (h : 0 < ) :
  |Real.log t| (t ^ + t ^ (-)) /

theorem log_fluctuation_integrableOn ( c a : ℝ) (h : 0 < ) (hc : 0 < c) :
  IntegrableOn (fun t => |Real.log t| * t^(c-1) * Real.exp (-
    *t + *a*Real.sqrt t)) (Set.Ioi 0)

theorem hasDerivAt_fluctuation_order ( lam a : ℝ) (h : 0 < ) (hlam : 0 < lam) :
  HasDerivAt (fun l => fluctuation l a)
    ( t in Set.Ioi 0, Real.log t * t^(lam-1) * Real.exp (-*t + *a*Real.sqrt t)) lam
```

3 Proof strategy

Mathlib’s parametric-differentiation lemma¹ needs: the pointwise λ -derivative of the integrand, a dominating function valid on a neighbourhood of λ , and its integrability.

¹`hasDerivAt_integral_of_dominated_loc_of_deriv_le`.

- **Pointwise.** $\partial_\lambda t^{\lambda-1} = t^{\lambda-1} \log t$ from `Real.hasStrictDerivAt_const_rpow` (derivative of $x \mapsto t^x$) composed with $\lambda \mapsto \lambda - 1$; multiply by the (λ -independent) Gaussian factor.
- **Domination.** On ball $\lambda(\lambda/2)$ every order $l \in (\lambda/2, 3\lambda/2)$ is positive, and $t^{l-1} \leq t^{\lambda/2-1} + t^{3\lambda/2-1}$ (monotone/antitone in the exponent according as $t \geq 1$). So $\|(\log t)t^{l-1}e^{\cdots}\| \leq |\log t|(t^{\lambda/2-1} + t^{3\lambda/2-1})e^{\cdots}$, a sum of two $|\log t|$ -weighted fluctuation integrands.
- **Log integrability.** For $c > 0$, $|\log t|t^{c-1}e^{-\beta t + \beta a\sqrt{t}}$ is integrable: bound $|\log t| \leq (t^\varepsilon + t^{-\varepsilon})/\varepsilon$ with $\varepsilon = c/2$ (`abs_log_le_rpow_add`, from `Real.log_le_rpow_div`), so the integrand is $\leq (1/\varepsilon)(t^{c-1+\varepsilon} + t^{c-1-\varepsilon})e^{-\beta t + \beta a\sqrt{t}}$; the AM–GM phase bound $\beta a\sqrt{t} \leq \frac{\beta}{2}t + \frac{\beta}{2}a^2$ reduces each term to `integrableOn_rpow_mul_exp_neg_mul_rpow` at $p = 1$ (exponents $c - 1 \pm \varepsilon > -1$).

4 Roadblocks and resolutions

- **The derivative integrand is signed.** $\log t$ is negative for $t < 1$, so $\|(\log t)t^{l-1}e^{\cdots}\|$ must be handled by `abs_mul` down to $|\log t| \cdot t^{l-1} \cdot e^{\cdots}$ (the last two factors positive, `abs_of_pos`) — *not* `abs_of_nonneg` on the whole product, which is false.
- **The $|\log t|$ bound is two-sided.** `Real.log_le_rpow_div` gives only the upper side; the lower is recovered by applying it to t^{-1} (`Real.log_inv`, `Real.inv_rpow+rpow_neg`), yielding the symmetric $t^\varepsilon + t^{-\varepsilon}$ bound valid on all of $(0, \infty)$.
- `Real.hasStrictDerivAt_const_rpow.hasDerivAt.comp` carries a `Module`-instance diamond; `simp only [Function.comp, mul_one]` at `h`; `exact h` lets `defeq` reconcile it (as in the earlier Weber unit).
- Exponent arithmetic $t^{c-1}t^{-(c/2)} = t^{c-1-c/2}$ needs the RHS exponent pre-normalised to $(c - 1) + (-(c/2))$ before `Real.rpow_add`, since $a - b$ and $a + (-b)$ are `defeq` but not syntactically equal (so `rpow_add` alone leaves the goal open).

5 What was Mathlib, what was new

Mathlib supplied the parametric-FTC lemma, `Real.log_le_rpow_div`, `Real.hasStrictDerivAt_const_rpow`, the `rpow` monotonicity lemmas, and `integrableOn_rpow_mul_exp_neg_mul_rpow`. New: the two-sided log–power bound, the $|\log t|$ -weighted fluctuation integrability (a reusable analytic lemma), and the order-derivative itself — the first differentiation of S_λ in its order rather than its argument.

6 Lessons

Differentiating in an exponent parameter is dominated the same way as in an ordinary parameter, but the log that drops out demands a $|\log|$ -integrability lemma Mathlib lacks; the clean route is the symmetric power sandwich $|\log t| \leq (t^\varepsilon + t^{-\varepsilon})/\varepsilon$, which turns the domination into two ordinary $t^q e^{-bt}$ integrals. Keep the derivative integrand’s sign in mind: its norm factorises through `abs_mul`, never `abs_of_nonneg`.

7 Follow-ups

With the order-derivative in hand, `prop:LogODE` (differentiate the Weber ODE k times in λ) and `lem:log_shift/lem:log_generating` (which reorganise ∂_λ^k) are within reach — they iterate this identity. The remaining ladder-operator and commutator statements still want a Weyl-algebra framework.